

2. Basic Mechanics

One-line summary

An observation-based, step-by-step organization of equipment-inheritance processing in RF4SP and RF5.

This chapter organizes the equipment-inheritance mechanics of RF4SP and RF5 on the basis of observations, focusing on how the individual processes may interact and affect the final arrangement result.

This chapter includes:

- observed results;
- provisional organization; and
- hypothesis-based inferences.

Many parts of the internal algorithm remain unanalyzed, and the interpretation may change as additional testing is conducted. In this document, the term **randomization** refers to changes in selection behavior that may involve internal random-number processing (RNG).

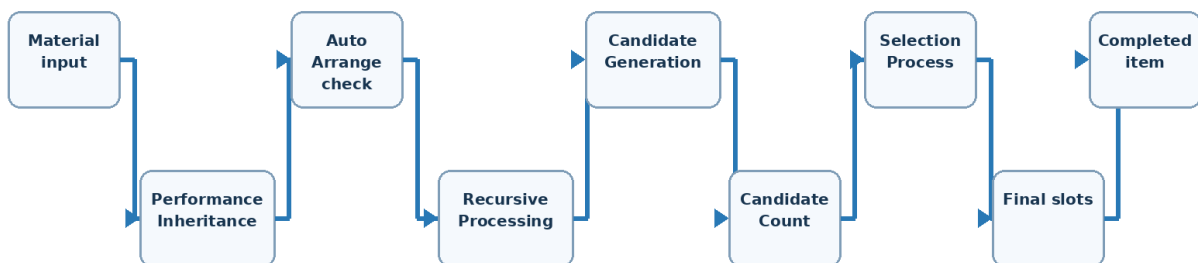
2-1. Overall Inheritance-Processing Flow

One-line summary

Inheritance proceeds through multiple stages, from material input to generation of the completed item; candidate generation and selection affect the probability of success.

Overall Inheritance-Processing Flow

Observation-based organization; individual stages may involve unknown internal transitions



Candidate Generation, Candidate Count, and Selection strongly affect practical success rates.

Figure 2-1. Observation-based organization of the full inheritance-processing flow.

At least within the scope of the present observations, inheritance processing in RF4SP and RF5 may consist of the following multiple stages:

Material input

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Performance Inheritance determination

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Auto Arrange occurrence determination

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Recursive Processing

↓
Candidate Generation
↓
Candidate Count determination
↓
Selection Process
↓
Final arrangement-slot determination
↓
Generation of the completed item

This document separates these stages into individual processes and organizes their relationships with one another.

In particular, the following are treated as important factors that substantially affect success probability and the expected number of attempts:

- Candidate Generation;
- Candidate Count; and
- the Selection Process.

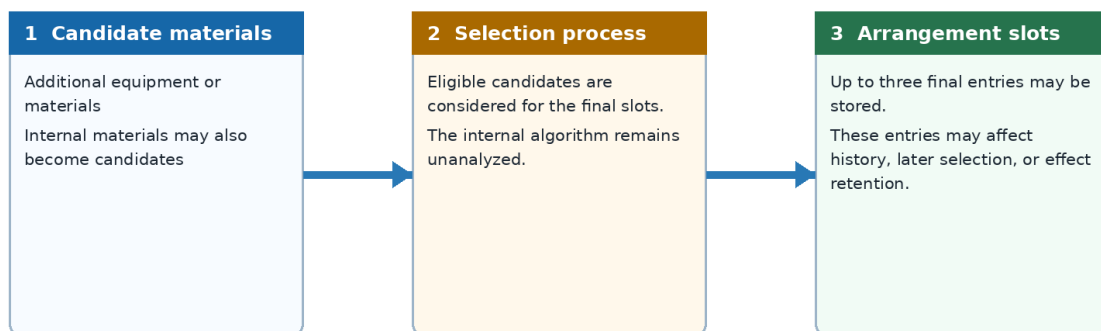
2-2. Separation of Performance Inheritance and Arrangement Inheritance

One-line summary

Performance Inheritance reflects stats or special effects in the item itself, while Arrangement Inheritance stores material information in arrangement slots; they are likely to be separate processes.

Arrangement Inheritance — Concept

Selected materials are stored as arrangement-slot entries



Observed boundary: RF4SP and RF5 may handle arrangement slots differently.

Figure 2-2. Arrangement-slot registration as a process separate from Performance Inheritance.

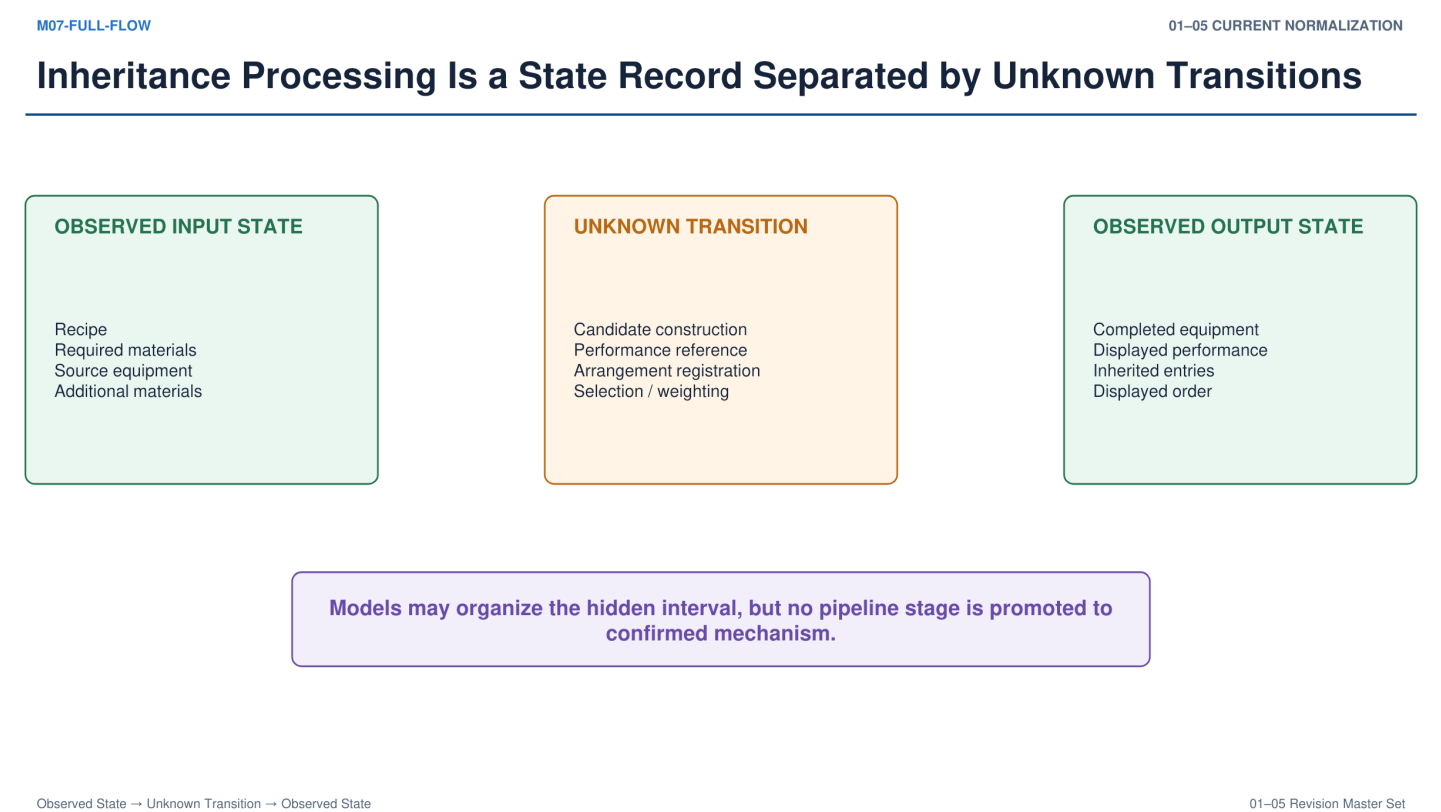
At least within the scope of the present observations, **Performance Inheritance** and **Arrangement Inheritance** may be handled as separate processes.

Performance Inheritance reflects the following in the completed item itself:

- numerical stats;
- elemental resistances; and
- special effects.

Figure 2-1

Placement 02-01 | Master M07-FULL-FLOW | Japanese Canonical p.2



Local caption
Full inheritance-processing flow

Current normalization: Observation, Model, and Unknown Transition remain explicitly separated.

Arrangement Inheritance, by contrast, registers the following as arrangement-slot entries:

- additional materials;
- internal materials; and
- inheritance history.

In RF5 in particular, the source item's special effects have been observed to remain even when the source item itself was excluded from the final arrangement slots.

Therefore, the following two conditions are not necessarily identical:

- being present in an arrangement slot; and
- having the performance reflected in the item itself.

2-3. Components of the Candidate Count

One-line summary

The Candidate Count during selection may consist of the performance source, ordinary materials, Auto Arrange materials, Performance-Source Contamination, and candidates produced through Recursive Processing; increases in the count reduce success probability.

Element	Classification	Source	Effect on Candidate Count	Notes
Performance source	Basic candidate	The first item selected	1	At least within the scope of the present observations, it may be included in the candidate pool.
Ordinary arrangement materials	Basic candidates	Additional equipment or materials added by the player	Increases by the number added	Become candidates for the final arrangement slots.
Auto Arrange materials	Automatically generated	When a recipe requires four or more materials and arrangement slots are unfilled	Automatic increase	Materials may be selected at random from the n required materials.
Performance-Source Contamination	Special phenomenon	The source equipment itself becomes a candidate again	Candidate increase	Observed in both RF4SP and RF5.
Candidates produced through Recursive Processing	Internal reference	Reference to the source equipment's internal arrangement entries	Candidate increase	Internal materials may become candidates.
Internal arrangement entries	Stored information	Already stored inside the source equipment	Context-dependent	May become candidates through Recursive Processing.
Candidate Generation	Phenomenon	Recursive Processing, Performance-Source Contamination, and similar processes	Candidate Count increase	Materials that were not originally added enter the candidate pool.
Candidate Count of four or more	Boundary condition	After candidates have been added	May be randomized	In RF5, reselection may occur.
Candidate Count of three or fewer	Boundary condition	No increase, or only a small candidate pool	Tends toward fixed behavior	Observed in RF5.
Selection Process	Decision process	After the candidate pool has been finalized	Determines the final slots	Up to three slots may be selected.
Final arrangement slots	Final result	Result of selection	Reduces the count	Only materials selected from the candidate pool are registered.

In this document, **Candidate Count** means the total number of eligible targets present when the Selection Process occurs.

At least within the scope of the present observations, the Candidate Count may be composed of:

- the performance source;

- ordinary arrangement materials;
- Auto Arrange materials;
- Performance-Source Contamination; and
- candidates produced through Recursive Processing.

In particular, the following may cause materials that the player did not directly add to enter the selection candidate pool:

- Performance-Source Contamination;
- Recursive Processing; and
- Candidate Generation.

An increase in Candidate Count has a substantial effect on:

- lower success probability;
- a greater expected number of attempts; and
- changes in the efficiency of split-save strategies.

Candidate Count is therefore treated as a key concept in this document.

2-4. Organized Differences Between RF4SP and RF5

One-line summary

RF4SP and RF5 likely show observable differences in selection randomness, the handling of special effects, and the composition of arrangement slots.

Item	RF4SP	RF5	Confidence	Notes
Initial selection behavior	May be strongly random from the first attempt	May tend toward fixed behavior when the Candidate Count is three or fewer	Medium	Stable ordering was observed in RF5, including tests using Object X.
Candidate Count of four or more	May tend toward random behavior at all times	Reselection or shuffling may occur	High	Observed in RF5 boundary-value tests.
Candidate Count of three or fewer	Random behavior may occur	Behavior tending toward fixed ordering was observed	High	Confirmed repeatedly in RF5.
Method of retaining special effects	May depend on arrangement slots	May become fixed to the item itself at the time of inheritance	Medium	In RF5, the effect was observed to remain even when the source was outside the slots.
When the source is excluded from the slots	Special effects may be lost	Special effects were observed to remain	High	Confirmed with Magic Ring and other items.
Numerical-stat inheritance	References the source performance	References the source performance	High	Confirmed for shoes and accessories in both games.
Shoe and accessory inheritance	References source performance rather than appearance	References source performance rather than appearance	High	Common observation in both games.
Special-effect inheritance through arrangement	Its existence was confirmed	Its existence was confirmed	High	Initially unconfirmed, but successful cases were later observed.
Four-entry arrangement-slot composition	Confirmed	Confirmed	Medium	Example: one-slot base weapon + source equipment (no arrangement entries) + Light Ore + three Gliitta Augites.

Item	RF4SP	RF5	Confidence	Notes
Performance-Source Contamination	Observed	Observed	High	Confirmed with accessories.
Candidate Generation through Recursive Processing	Observed	Observed	Medium	Derived from references to internal information in the source equipment.
Internal Candidate Generation from additional arrangement materials	Unconfirmed	Unconfirmed	Medium	Could not be confirmed at present.
Auto Arrange	Observed	Observed	High	May occur when a recipe requires four or more materials.
Auto Arrange selection	May be random	May be random	Medium	Inclusion of the first material was confirmed.
After completion of Auto Arrange	May become an ordinary arrangement-slot entry	May become an ordinary arrangement-slot entry	Medium	No special distinction has been confirmed.
Maximum number of selected slots	May be a maximum of three	May be a maximum of three	High	At least within the scope of the present observations.
State of internal-mechanics analysis	Many unknowns	Many unknowns	High	Common to both games.
Stability of the current understanding	Long history of research	Many recent discoveries	Low	The volume of community research differs.

At least within the scope of the present observations, differences may exist between the inheritance mechanics of RF4SP and RF5.

The principal candidate differences identified are as follows.

RF4SP:

- randomness may be strong from the initial selection;
- special effects may depend on arrangement slots; and
- a four-entry arrangement-slot composition was unconfirmed.

RF5:

- behavior may tend toward fixed ordering when the Candidate Count is three or fewer;
- reselection may occur when the Candidate Count is four or more;
- special effects may become fixed to the item itself; and
- a four-entry arrangement-slot composition was confirmed.

However, many possible differences in the internal algorithms remain untested, and additional testing is required.

2-5. Boundary Values

One-line summary

Inheritance processing may have boundary values involving the Candidate Count and the number of required materials; selection behavior may change when a threshold is crossed.

Auto Arrange Boundary Test

Observation-Based Boundary Condition Model

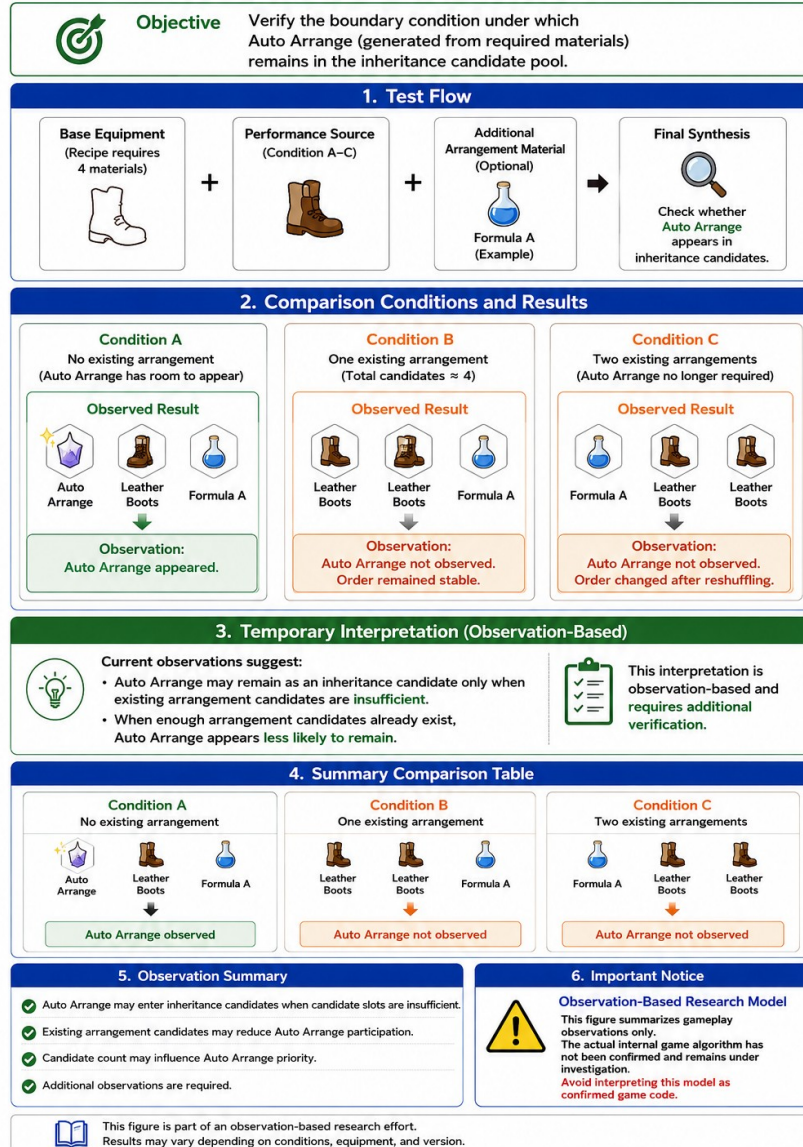


Figure 2-3. Boundary-test reference for candidate-count and Auto Arrange behavior.

At least within the scope of the present observations, inheritance processing may contain several boundary values.

Representative examples follow.

Candidate Count Boundary

When:

$$\text{Candidate Count} \leq 3,$$

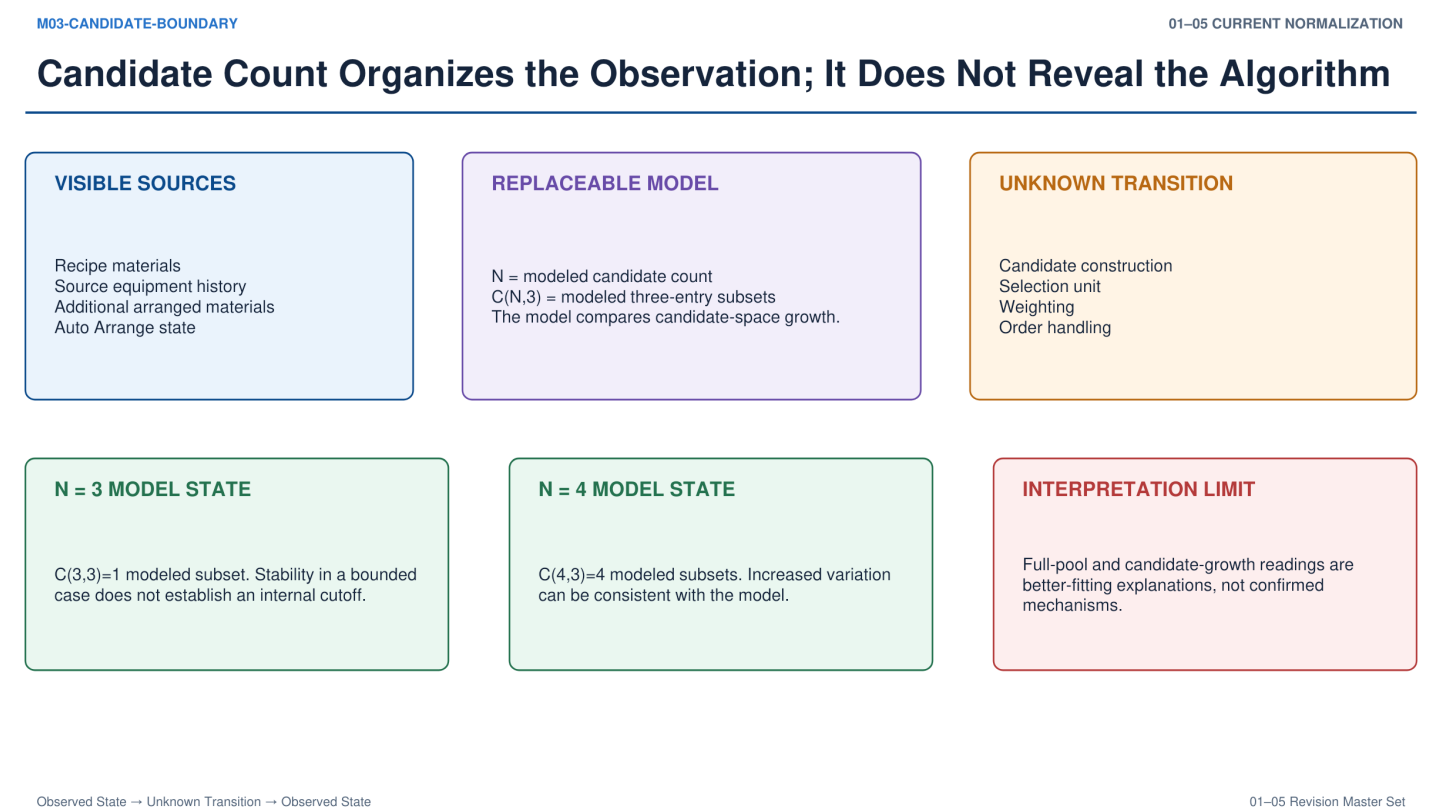
behavior tending toward fixed ordering was observed in RF5.

By contrast, when:

$$\text{Candidate Count} \geq 4,$$

reselection or randomization may occur.

Figure 2-3



Auto Arrange Boundary

Let the number of required materials be n .

When:

$$n > 3,$$

Auto Arrange may occur for the number of unfilled arrangement slots given by:

$$n - 3.$$

This document treats these boundary values as important indicators in the analysis of the inheritance mechanics.

2-6. Provisional Organization of Hypotheses

One-line summary

This chapter organizes observed behavior and high-confidence hypotheses while explicitly identifying unresolved questions such as selection weights and recursion depth.

Classification	Content	Confidence	Notes
Observed	Randomization with a Candidate Count of four or more in RF5	High	Boundary-value tests
High-confidence hypothesis	Separation of Performance Inheritance and Arrangement Inheritance	Medium	Differences in special-effect retention
Unresolved	Selection weights	Low	RNG unknown

This document distinguishes among:

- observed behavior;
- high-confidence hypotheses; and
- unresolved questions.

At present, the following are regarded as relatively high-confidence findings or interpretations:

- randomization with a Candidate Count of four or more in RF5;
- Performance-Source Contamination;
- Candidate Generation through Recursive Processing; and
- the possible separation of Performance Inheritance and Arrangement Inheritance.

By contrast, the following are treated as unresolved questions:

- selection weights;
- internal priority ordering;
- limits on recursion depth; and
- details of the internal algorithm.

Accordingly, the organization, terminology, and hypotheses presented in this document may be updated or revised through additional testing.